Western Blotting

M12 | ver 1.1 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Detect proteins in cultured-cell lysates by SDS-PAGE, wet transfer to PVDF and HRP chemiluminescence. Use intact membranes for sequential target and loading-control detection, with stripping between rounds.

Materials & Equipment

Material / equipment	Supplier	Specification / use
PBS, RIPA and SDS sample buffer	Laboratory stock	Cell washing, lysis and 1× final sample buffer
Precast gel and matched running buffer	Servicebio	12% gel used in the source workflow
Prestained protein ladder	Laboratory stock	Covers the expected target sizes
PVDF membrane	0.45 μm membrane	Protein immobilization
Absolute ethanol	Sangon Biotech	Membrane wetting
Wet-transfer buffer and consumables	Servicebio / laboratory stock	Compatible with the wet-transfer apparatus
Non-fat dry milk and TBST	Laboratory stock	5% (w/v) milk for blocking; TBST washes
Target primaries, HRP secondaries and HRP–β-actin	CST	Assignments and dilutions below
ECL reagents A and B	Servicebio	Mix 1:1 immediately before detection
Membrane stripping reagent	Laboratory stock	Compatible with PVDF and repeat immunodetection
Refrigerated centrifuge, heating block, electrophoresis and wet-transfer apparatus	Eppendorf centrifuge; laboratory equipment	Lysis, denaturation and separation
Ice bath, rocker and chemiluminescence imager	Laboratory equipment	Cooling, incubation and unsaturated imaging

Before You Start

Plan matched lane order, a molecular-weight ladder and appropriate positive/negative controls. Use running and transfer buffers compatible with the installed gel/apparatus.

Prepare 5% milk/TBST (5 g per 100 mL final volume). Select antibody-compatible incubation diluent; the documented settings apply to this team workflow. [1]

Procedure

A. Prepare cell lysates

1. Remove culture medium, wash cells with PBS and collect into microcentrifuge tubes. Remove residual PBS. For a 12-well culture, resuspend the collected cells in 100 µL RIPA and incubate on ice for 20-30 min. [1]

2. Centrifuge at 12,000 × g for 10 min at 4°C. Transfer the clear supernatant into a clean tube without disturbing the pellet. Keep lysates cold.

3. For protein-abundance comparisons, determine protein concentration with a RIPA-compatible assay and normalize the protein input across samples. Keep final loading volumes comparable; equal lysate volume alone does not establish equal protein loading.

4. Add SDS sample buffer to 1× final concentration. For an n× stock, add Vsample/(n - 1) of buffer. Heat at 95°C for 10 min, cool and briefly centrifuge. Use a different denaturation condition only when required for the target. [1]

B. Separate and transfer proteins

5. Install a 12% precast gel and add its matched running buffer. Rinse the wells and remove loose gel fragments without damaging the well bottoms. [1]

6. Load a molecular-weight ladder and the prepared samples according to the lane map. The source run used 6 µL prepared sample per lane; select a load compatible with the well capacity and the validated detection range. [1]

7. Run at 120 V with ice-water cooling. Monitor the tracking dye and stop when separation is sufficient, before proteins of interest leave the gel. The source run used the dye front reaching the gel exit as its endpoint. [1]

8. Wet a 0.45 µm PVDF membrane in absolute ethanol for 2 min, then equilibrate it in transfer buffer. Keep the membrane wet during assembly. [1]

9. Assemble the wetted sandwich with the gel on the cathode side and PVDF on the anode side. Remove all bubbles between gel and membrane, close the cassette and load it in the correct orientation.

10. Wet-transfer at constant 400 mA for 70 min in an ice-water bath. Record the number of gels in the apparatus; do not multiply this setting by the number of gels without validating the setup. Retain the ladder image and check transfer before immunodetection. [1]

C. Block and detect the target

11. Block the intact membrane in 5% milk/TBST for 1 h with gentle agitation. Wash with fresh TBST three times, 5 min per wash.

12. Incubate with the selected target primary antibody at 1:1,000 overnight at 4°C. Use sufficient solution to cover the membrane and maintain gentle agitation. Antibody assignments from the source workflow are shown below. [1]

Target	Primary host	Secondary
His-tagged protein	Rabbit	Anti-rabbit HRP
FLAG-tagged protein	Mouse	Anti-mouse HRP
Total APOE	Rabbit	Anti-rabbit HRP
APOE4	Rabbit	Anti-rabbit HRP

13. Wash three times with TBST for 5 min each. Incubate with the matching HRP secondary at 1:3,000 for 1-2 h at room temperature. Wash again with fresh TBST three times, 5 min each.

14. Mix Servicebio ECL reagents A and B at 1:1 immediately before use. Prepare enough solution to cover the membrane, protect it from light and apply for the selected substrate's specified development time. [1]

15. Acquire a range of exposures and retain unsaturated images with resolved bands. Save raw target and ladder images before stripping; select exposure by signal quality rather than reusing fixed seconds from another experiment.

D. Strip and reprobe the intact membrane

16. Wash away residual ECL and fully immerse the membrane in the selected PVDF-compatible stripping reagent. Incubate at room temperature for 10 min.

17. Remove stripping reagent and wash thoroughly according to its product instructions. Check removal of prior detection reagents before adding the next antibody: fresh ECL checks residual active HRP, while a secondary-only test can reveal residual primary antibody when an indirect detection round follows. [3]

18. Re-block the membrane in 5% milk/TBST for 1 h. Incubate with directly HRP-conjugated anti-β-actin at 1:3,000 for 1 h at room temperature. Do not add a secondary antibody for this directly conjugated reagent. [1]

19. Wash with TBST three times for 5 min each and detect with freshly mixed ECL. Preserve lane identity and pair the target image with the loading-control image from this same membrane. For further targets, repeat stripping and reblocking only after checking that signal recovery remains suitable.

E. Archive and analyze

20. Archive uncropped images, exposures, lane map, antibody identities, dilutions and stripping history. For densitometry, use unsaturated bands, consistent background subtraction and the matched loading control. Report independent biological replicates separately from repeated exposures of one membrane.

p.s.

Keep all membranes intact during immunodetection. Mark orientation outside the analytical region so lane identities remain traceable after successive washes and stripping.

When preparing lysates, use appropriate protease protection for the selected RIPA reagent and target. Keep samples cold before denaturation and avoid repeated freeze-thaw cycles. [4]

Change the wash solution each time and keep membranes submerged with gentle movement. Dry patches and trapped bubbles can create uneven signals.

Stripping can remove protein as well as antibody. Use the same detection order across comparable samples and investigate signal loss rather than assuming every later round is equivalent to the first. [2]

Expected Outcome

Obtain interpretable bands near the anticipated molecular weight, acceptable background and unsaturated target/loading-control images. Assess specificity with appropriate positive and negative controls. Protein abundance or an isoform-reactive band alone does not establish an RNA-editing rate.

Troubleshooting

Observation	Check first	Action
Weak or absent target signal	Positive control, input, transfer, antibody host and detection reagents.	Verify successful transfer and use fresh correctly paired antibodies/ECL. Check sample integrity before increasing exposure or input. [4]
High background or saturated bands	Wash completeness, antibody dilution, blocker and exposure.	Repeat washing and shorten exposure. Titrate antibody when needed; do not quantify clipped signals. [4]
Blank patches or uneven bands	Bubbles, membrane wetting and solution coverage.	Remove gel-membrane bubbles during assembly and keep the membrane uniformly immersed. Repeat the blot if the analytical region was not transferred.
Smearing or distorted migration	Overloading, viscous lysate, sample preparation and heating.	Review input and clarification; reduce excessive loading and maintain cooling. Verify gel/buffer compatibility.
Previous bands remain after stripping	Residual HRP or primary antibody; stripping history.	Check stripping efficiency before reprobing. Follow reagent-specific limits; use a fresh matched blot if removal remains incomplete. [2, 3]
Signal diminishes after repeated probing	Protein loss or epitope damage during stripping.	Minimize additional stripping and compare against a fresh blot. Do not treat a weaker later-round signal as a biological change. [2]
Loading-control bands vary strongly	Protein input, transfer, exposure saturation and stripping effects.	Review all four factors before normalization. Repeat with comparable protein loading and a validated loading control if needed.

Reference

[1] SUSTech-BIO 2026. WB record(1).docx, supplied team experimental record, sections 3-4; supplemented and corrected by the user on 2026-09-27. Local source retained in sources/.

[2] Cell Signaling Technology. Do you have a recommended protocol for stripping/re-probing blots? Updated 2026-06-30; accessed 2026-09-27. Open source

[3] Thermo Fisher Scientific. Technical Tip: Strip and reprobe western blots, pp. 1-2. General stripping-efficiency checks; accessed 2026-09-27. Open source

[4] Cell Signaling Technology. Western Blotting Troubleshooting Guide. Accessed 2026-09-27. Open source